

COLLEGE PROJECT REVIEW

Portfolio Risk-Factor Analysis Using PCA

Dimensionality Reduction for Portfolio Risk Management

Team Members

B Shammruth

U. Praneeth

Problem Statement

1

Too many risk factors

A portfolio of N assets has $N(N-1)/2$ correlations. Risk becomes hard to see and manage.

2

Assets move together

Stocks in a market move in the same direction, so holding many assets may not diversify as much as expected.

3

Noisy covariance matrix

With limited history, large covariance matrices are unstable, which hurts optimisation and VaR estimates.

4

Unclear sources of risk

Managers need to know what drives risk (market, sector), not just how big it is.

66

pairwise correlations for just 12
stocks

78

covariance entries to estimate. Goal:
explain them with 2 to 4 factors

Objectives & Proposed Solution

Objectives

- Find the main hidden risk factors
- Reduce many correlated assets to few components
- Measure how much risk each factor explains
- Understand what each factor means (market, sector)
- Support better diversification and hedging

Expected outcome: a simple, explainable view of portfolio risk.

Proposed Solution

Asset returns

12 stocks, daily



PCA

Eigen-decomposition of correlation matrix



Key risk factors

PC1, PC2, PC3 ...



Risk decisions

Diversify, hedge, monitor

Dataset & Exploratory Data Analysis

12

stocks

750

trading days (~3 yrs)

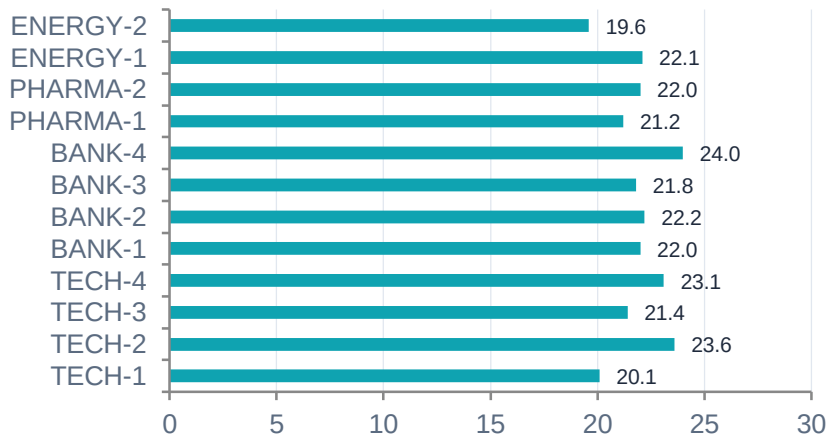
4

sectors

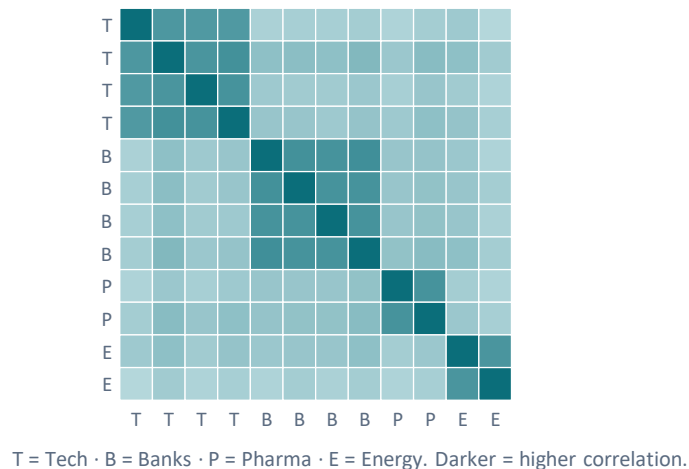
0.81 vs 0.54

corr. same vs different sector

Annualised Volatility (%)



Correlation heatmap (12 stocks)



Key Concepts Behind PCA

1

Variance = Risk

Volatility comes from variance of returns. PCA finds the directions where variance is largest.

2

Eigenvector = Loading

Weights of each stock in a component. Shows what the factor represents.

3

Eigenvalue = Size

Amount of variance a component captures. Larger eigenvalue means a more important risk factor.

4

PC Score = Factor return

Daily value of the component, $PC = Z \cdot v$. Components are uncorrelated with each other.

$$\text{Explained variance of } PC_k = \lambda_k / (\lambda_1 + \lambda_2 + \dots + \lambda_N)$$

Practical Implementation: Program Workflow



Input: 12 x 750 daily returns **Output:** eigenvalues, loadings, PC scores, risk share per factor

Environment: Python 3 · NumPy · pandas · scikit-learn · Matplotlib

PCA Algorithm & Implementation

1

Standardise returns

$$Z = (R - \text{mean}) / \text{std}$$

2

Correlation matrix

$$\Sigma = Z^T Z / (n - 1)$$

3

Eigen-decomposition

$$\Sigma v = \lambda v$$

4

Rank by eigenvalue

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N$$

5

Project data

$$PC = Z \cdot v$$

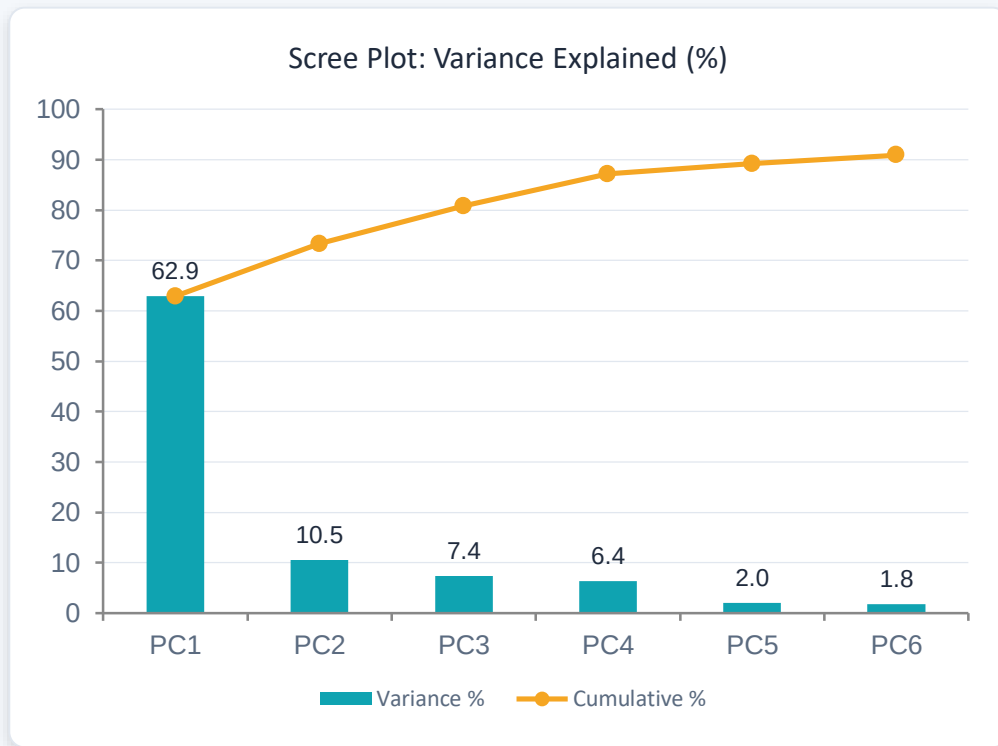
```
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

Z = StandardScaler().fit_transform(R)
pca = PCA().fit(Z)
ratio = pca.explained_variance_ratio_
loads = pca.components_      # loadings
scores = pca.transform(Z)
```

How many components to keep?

Cumulative variance $\geq 85\text{--}90\%$, or eigenvalue > 1 (Kaiser rule). Here Kaiser keeps 2 components.

Results 1: Risk Explained by Each Factor

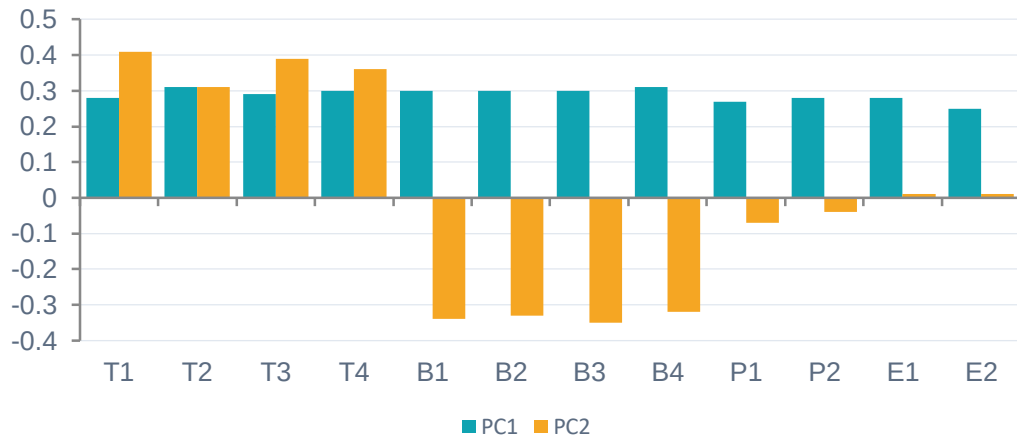


PC	Eigenvalue	Var %	Cum %
PC1	7.54	62.9	62.9
PC2	1.26	10.5	73.3
PC3	0.89	7.4	80.8
PC4	0.77	6.4	87.2

- PC1 alone explains 62.9% of total risk
- Sharp drop after PC2: the elbow
- 4 PCs capture 87.2%; 6 reach 90%
- Only PC1, PC2 have eigenvalue > 1

Results 2: What Do the Risk Factors Mean?

Loadings of PC1 and PC2



T = Tech · B = Banks · P = Pharma · E = Energy

PC1: Market factor

All loadings positive (~ 0.3). Correlation with market = 0.92.

PC2: Tech vs Banks

Tech positive, Banks negative: a sector rotation factor.

$\approx 100\%$ of equal-weight portfolio risk comes from PC1

Risk insight: diversifying across similar stocks does not remove market risk. Hedge PC1, and use PC2 to manage sector tilt.

Conclusion, Future Scope & References

Conclusion

- PCA reduced 12 correlated stocks to a few key factors
- PC1 = market factor (62.9% of variance)
- PC2 = Tech vs Banks sector factor
- 4 PCs keep 87.2% of information
- Gives a clearer, more stable view of risk

Future Scope

- Test on real NSE / global data
- Rolling-window PCA to track changing factors
- Link factors to VaR, stress tests, hedging
- Explore Sparse PCA and statistical factor models
- Limitation: PCA is linear and factors need interpretation

References

- Jolliffe & Cadima (2016), Principal component analysis: a review and recent developments. Phil. Trans. R. Soc. A
- Markowitz (1952), Portfolio Selection. Journal of Finance
- Alexander (2008), Market Risk Analysis, Vol. II. Wiley